

Dear Editor,

We are pleased to submit our manuscript entitled “**NestNetR: Deep-learning based analysis of breeding behaviour from light and temperature recordings in R**” for consideration for publication in *Ornithological Applications*.

Biologging is rapidly expanding our ability to study avian behaviour, but the pace and volume of time-series data now exceed what can be processed consistently with manual or ad hoc approaches. As a result, there is an increasing need for objective, scalable, and reproducible workflows that convert biollogger recordings into comparable behavioural metrics. This need is particularly pronounced in breeding ecology, where robust inference on incubation, brooding, and breeding phenology underpins population-level analyses and conservation decisions, especially in remote regions with limited field access.

In this study, we introduce **NestNetR**, an open-source R package that provides a fully integrated workflow for behavioural classification from light-level geolocator data using deep learning. NestNetR supports data import and preprocessing, model use or custom model definition, and automated daily classification of breeding-related behaviour within a modular, reproducible framework implemented entirely in R. Using an extensive Arctic breeding-season dataset of shorebirds, we demonstrate that breeding behaviours relevant to reproductive success can be inferred accurately from combined light and temperature time series, and that trained models can be transferred across species.

We believe this manuscript aligns well with the scope of *Ornithological Applications* by providing a practical analytical tool that enables the extraction of biologically meaningful breeding information from widely used avian biologging devices.

We confirm that this manuscript has not been submitted elsewhere, that all authors have approved the manuscript in its current form, and that there are no conflicts of interest to declare.

Thank you very much for your consideration.

Kind regards,

Bennett Paul Stolze

(on behalf of all authors)